ELSEVIER

Combustion Aerodynamics of a Gas-Fired Furnace with Peripheral Fuel Injection

A. M. A. Kenbar
S. A. Beltagui
The University of Glasgow and National Engineering Laboratory (NEL), East Kilbride, Glasgow, United Kingdom

N. R. L. Maccallum
The University of Glasgow, Glasgow, United Kingdom

■ As part of a research program to study the combustion characteristics of swirling flames in furnace systems, studies are aimed at improving combustion performance with reduced pollutant emissions. In addition to providing more insight into the processes involved—aerodynamics, combustion, heat transfer, and formation of pollutants—results of these studies are also needed for the validation of furnace prediction models. Measured flow and combustion patterns carried out on a semiindustrial-scale natural-gas-fired furnace are presented. The burner uses a nonconventional fuel injection scheme with the fuel injected around the periphery of the swirling air jet. The furnace is a water-cooled cylindrical combustion chamber of 1-m diameter and 3-m height, fired by natural gas through a variable swirl burner and a quarl. Measurements of the flow and combustion patterns were carried out for two air swirl intensities, swirl numbers 0.9 and 2.25, through radial traverses at 13 axial planes along the furnace. The flow patterns are defined by the radial distributions of the three time-averaged velocity components and static pressure. Also presented are the combustion patterns in the form of measured contours of temperatures and species concentrations of O_2, CO_2, CO, HC, and NO_x. The results demonstrate that peripheral fuel injection produces high rates of mixing, leading to better combustion efficiency and heat transfer. The axial velocity profiles define the main shear areas and the forward- and reverse-flow zones. The temperature and concentration fields illustrate the progress of combustion reactions to completion and the formation of pollutants. The data obtained by the detailed measurements are being used for the validation and development of mathematical models for prediction of furnace flows.

Keywords: *furnace, swirling flame, natural gas, peripheral fuel injection, combustion-generated NO_x, combustion aerodynamics, combustion patterns*

INTRODUCTION

Furnace system design requires careful characterization and optimization of complex flow and combustion phenomena. Although computational fluid dynamic (CFD) techniques are being progressively used in furnace design, there is still a great need for experimental measurements to test new designs and validate CFD predictions. Thus a mixed approach of physical model testing and CFD simulation reduces the time and cost of development. This approach is being followed in a research program to study the combustion characteristics of swirling flames in furnaces. The program is aimed at improving combustion performance with reduced pollutant emissions.

In nonpremixed industrial burners, mixing occurs within the combustion chamber by shear layer mixing between the air and fuel streams. Swirling the airflow can be used to increase shear mixing and enhance it with centrifugal mixing. The realization of the full benefits of these two physical processes in a practical combustion system depends mainly on the method of fuel injection.

Shear layer mixing is effectively utilized if the fuel is injected into the regions of maximum shear in the airstream. Centrifugal mixing effects can be used by creating favorable density gradients within the flow of air, fuel, and products. The most common method of fuel injection is axially at the center of the burner airflow. This method does not realize the full potential of either shear layer or centrifugal mixing.

As an alternative to central fuel injection, the fuel can be injected at the periphery of the entering air jet [1, 2]. This system offers better shear layer and centrifugal mixing, thus resulting in efficient mixing and hence high combustion intensity. Flame stability was achieved through

Address correspondence to Dr. S. A. Beltagui, Heat Transfer Unit, N.E.L. Executive Agency, East Kilbride, Glasgow G75 0QU, United Kingdom.

Experimental Thermal and Fluid Science 1995; 10:335–346
© Elsevier Science Inc., 1995
655 Avenue of the Americas, New York, NY 10010

0894-1777/95/$9.50
SSDI 0894-1777(94)00081-I

either the central or outer reverse flow of hot products. This system and similar systems have been examined by a number of investigators whose work is reviewed in [1].

The peripheral fuel injection system was studied [1] in an adiabatic furnace of 0.225-m inside diameter and 0.9-m length. This system resulted in a stable flame even without a central reverse-flow zone (CRZ); thus only low swirl settings, up to that which formed a CRZ, were used. Compared to the central axial fuel injection, the peripheral injection produced flames of higher intensity with wider stability ranges. This work gave strong evidence in support of fuel injection at the outer periphery of the swirled airflow. However, these findings are related to a small-scale adiabatic furnace where the flame stability is easier to achieve than in the large nonadiabatic furnace systems that exist in industry. It is necessary to test this scheme in a nonadiabatic industrial-size furnace system. Thus the present work was carried out on the NEL test facility, which represents a semiindustrial-scale furnace system. The main objectives of the NEL program were the following.

1. To acquire data to study different fuel injection methods in a semiindustrial-scale application. This should provide more insight into the processes involved, that is, aerodynamics, combustion, heat transfer, and formation of pollutants. The information is needed to optimize both the fuel injection mode and swirl to achieve the best combustion performance with minimum pollutant formation. In this paper detailed measurements of the flow and combustion patterns using the peripheral fuel injection scheme are presented. Comparisons with other fuel injection systems are given in [3].
2. To provide an extensive databank for use in developing and testing the theoretical programs for furnace flow and combustion. These include both furnace-specific codes such as PCOC [4] and more general CFD codes such as PHOENICS [5].

EXPERIMENTAL PROGRAM

NEL Furnace System

The NEL furnace models an upshot fired heater. The furnace is a water-cooled cylindrical combustion chamber of 1-m diameter and 3-m height. The furnace is fired by natural gas through a variable-swirl burner with a quarl. The burner uses a moving-block swirl generator. Two air swirl settings were tested with swirl numbers $S = 0.9$ and 2.25 to provide a wide range of data. The swirl number is defined as the ratio of tangential to axial momentum fluxes divided by the burner throat radius. More details of the furnace and measuring equipment are given in Refs. 6 and 7.

Entry Flow Arrangements

Fuel Entry Arrangements The fuel was injected axially at the outer periphery of the swirled airstream (Fig. 1). Two alternative arrangements were used. In arrangement A, the fuel entered through an annular slit of 2.5-mm width around the periphery of the swirled air. The velocity of the gas leaving the slit was 12 m/s. In arrangement B, the fuel was injected through 60 holes, each 4 mm in diameter. These holes were designed to maintain the same flow area as the continuous slit of arrangement A.

Both arrangements, A and B, produced flames of virtually identical geometry and appearance. Thus detailed measurements were taken for arrangement A only.

Air Entry Schemes In scheme 1 all the air entered through the burner throat (Fig. 1). It was found that a stable flame could be achieved only if enough swirl ($S > 0.8$) was applied to the airflow to produce a CRZ near the burner. However, of the two swirl settings investigated above this value, the one close to the stability limit ($S = 0.9$) produced a flame characterized by some nonuniformity in shape compared with the higher swirl flame ($S = 2.25$).

An alternative way of achieving the CRZ is to create an aerodynamic blockage by introducing part of the combustion air radially outward through a central gun. The gun used (Fig. 1) has 16 holes, each 5 mm in diameter, spaced on its outer periphery. Thus in scheme 2 about 10% of the combustion air was supplied through the central gun. In this case a stable flame was achieved even without swirl.

Measurement Conditions

Experimental measurements were performed to produce complete mapping of the flow and combustion patterns within the furnace. These measurements were carried out for the air entry of scheme 1.

The experimental measurements for the aerodynamic patterns include the three time-averaged velocity components and static pressure. Those for the combustion patterns covered temperature and species concentrations of HC, CO, CO_2, O_2, and NO_x. These measurements were taken under fixed firing conditions of 400 kW and 5% excess air.

MEASUREMENTS

Radial traverses were carried out at 13 planes along the furnace that were closely spaced near the burner. Some extended traverses were made to check for flow symmetry and intrusive effects of probes. The probe effects were found to be minor and the flow symmetry very reasonable. Repeatability tests gave velocity values repeatable within ± 0.5 m/s. Between 13 and 26 measurement points were used in each plane depending on the gradient of the variable being measured. The absolute positioning of the probe relative to the furnace wall and base plate was achieved to within ± 2 mm.

The gas temperature was measured using a suction pyrometer [8], which has an overall shield diameter of 15 mm and uses an S-type thermocouple. The suction rate corresponds to a gas velocity of 150 m/s at the hot junction. The measured probe efficiency [8] was found to be 99%; thus temperature measurements needed no correction. Temperature measurements were repeatable to within ± 10 K.

A spherical head five-hole Pitot probe of 8-mm tip diameter [7, 8] was used for the measurements of the three time-averaged velocity components and static pressure. The probe was recalibrated in a special rig before the start of the measurements [7].

Although the probe is intrusive, it has been confirmed

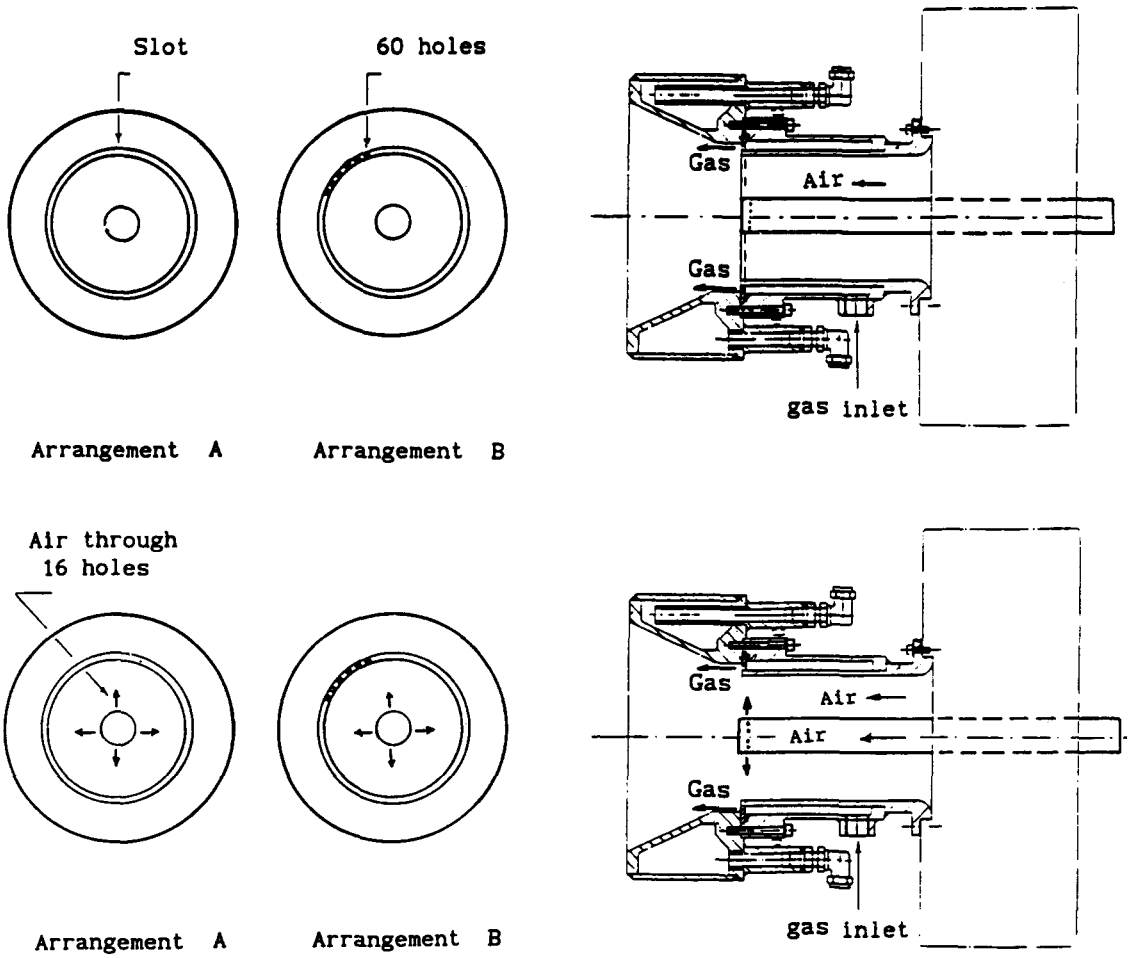


Figure 1. Air and fuel entry arrangements: Scheme 1 (*top*) and Scheme 2 (*bottom*).

in the present furnace that the velocity values obtained generally agree with those measured by laser-Doppler anemometry (LDA). The discrepancies observed typically never exceeded ± 1 m/s, the largest discrepancies occurring at the peak forward velocities (e.g., about 20 m/s) and in some of the reverse-flow zones. In some of the reverse-flow zones where LDA indicated a reverse flow of 2 m/s, the five-hole Pitot probe could indicate reverse flow of only 1 m/s. Over much of the forward flow field the discrepancy was only 0.5 m/s or less. Similar comparisons elsewhere reached the same conclusion (e.g. [9]). It is only in regions of low axial velocity and relatively high radial velocity that discrepancies occur between measurements by the two methods. There is therefore reasonable confidence in the axial and tangential velocity component measurements but less confidence in the radial component values.

Mass integration of the measured axial velocity distribution at any section compared well with the metered flow values, within 25%.

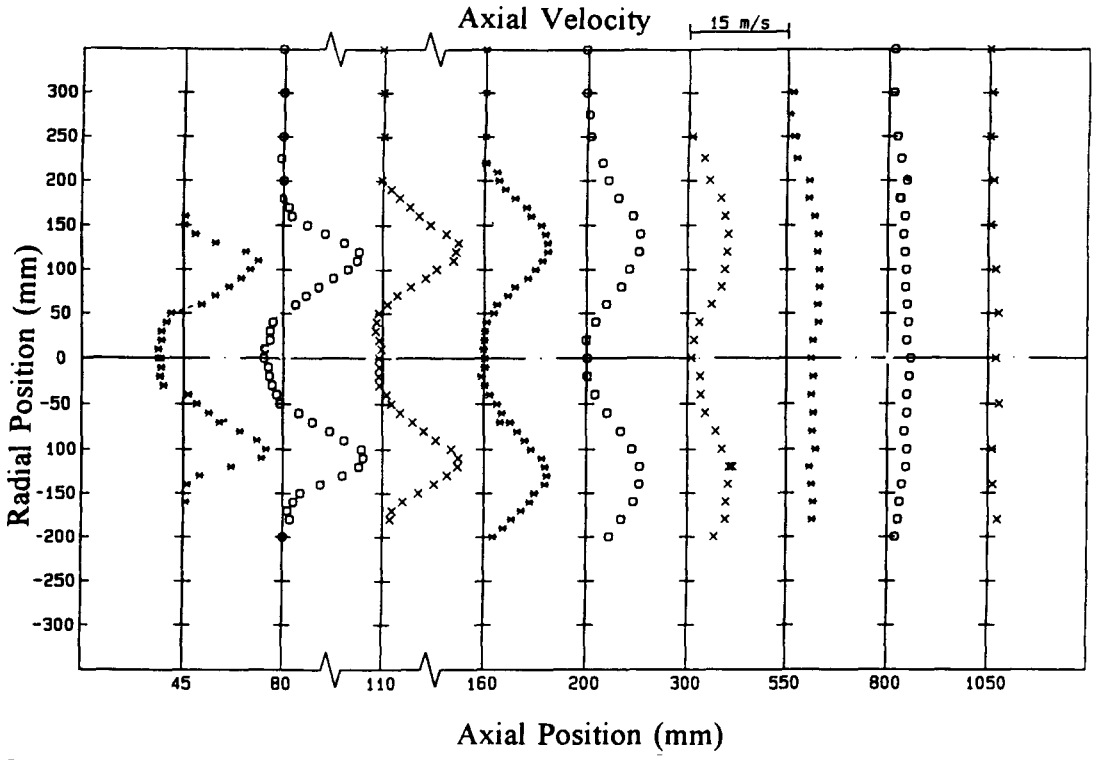
For concentration measurements, a special water-cooled stainless steel sampling probe [6] was used. For NO_x and HC analysis, the gas sample passed through a heated sample line to the NO_x chemiluminescent and hydrocarbon flame ionization detector analyzers, respectively. For CO, CO_2, and O_2 analysis, the sample was passed through a water trap, condenser, and chemical dryers before being introduced to the analyzers. Infrared absorption analyzers were used for CO and CO_2. A paramagnetic analyzer measured the concentration of O_2. Each analyzer was calibrated against known concentrations of the gas to be measured. The time-mean values of all measured variables are those averaged over a period of 1 min after sufficient time had been allowed for purging the sampling lines and analyzers. The concentration measurements were checked for repeatability, and the maximum deviation was within $\pm 1\%$ of the full scale for each analyzer.

FLOW PATTERN RESULTS

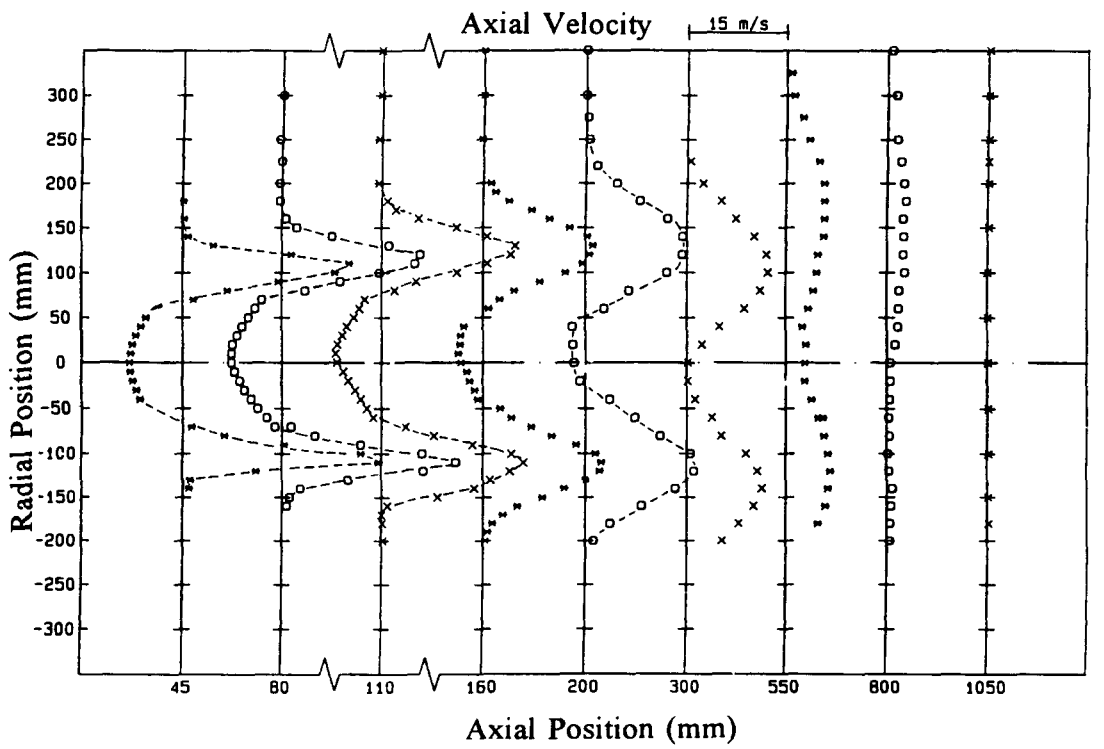
Flow pattern results are defined by the measured profiles of the three time-averaged velocity components and static pressure.

Axial Velocity Profiles

Axial velocity profiles are given in Fig. 2. The forward- and reverse-flow boundaries in each plane are defined by these profiles. For the two swirls tested, the flow patterns are essentially the same—type D, according to the classification of Beltagui and Maccallum [10]. In this flow pattern, near the burner the flow consists of CRZ surrounded by an annular jet containing the main forward flow. Outside the forward flow a weak external reverse-flow



a



b

Figure 2. Axial velocity profiles: (*a*) $S = 0.90$, (*b*) $S = 2.25$.

zone (ERZ) extended to the walls. The very low velocities in the ERZ observed here are due to the low confinement, the furnace-to-quarl diameter ratio being 5.0.

For the forward flow, the value of peak velocity increases with increased swirl. The high-velocity gradients at the boundaries of the forward flow increase even further with increased swirl. These gradients enhance the shear forces at the jet boundaries, leading to higher mixing and combustion rates. These features result in the short intense flames associated with the high swirl flows.

The rate of decay of maximum velocity along the furnace is accelerated at high swirl, this being associated with the jet area expansion (see next section). The recovery of the centerline axial velocity is enhanced with swirl.

Flow Boundaries

Figure 3 defines the boundaries of the forward flow and both the central and outer reverse-flow for the two swirls. For the first 250-mm distance from the quarl exit, the jet radial expansion is nearly the same for both swirls. Further downstream, the jet radial expansion increases with swirl, thus bringing the jet impingement point with the furnace wall nearer to the burner. It is noted that both the length and maximum diameter of the CRZ increase with swirl.

Tangential Velocity Profiles

The tangential velocity profiles are presented in Fig. 4. The tangential velocity values are an indication of the local swirl strength, which contributes to mixing and combustion. The high swirl jet approximates to the usual Rankine-vortex flow with solid-body central rotation surrounded by an outer free vortex. At planes from the furnace inlet to about 300 mm downstream, the tangential velocity values in the higher swirl are about twice the corresponding values of the lower swirl. However, the rate of decay of the tangential velocity increases with swirl, so the tangential velocities in both swirls decay to very low values by about 800 mm downstream.

Radial Velocity Profiles

The radial velocity profiles are similar for both swirl settings although the magnitudes of radial velocities are higher at the higher swirl. The profiles for the higher swirl ($S = 2.25$) only are shown in Fig. 5. In the upper half of the diagram, positive values correspond to outward radial velocity components, with the reverse convention in the lower half.

As mentioned in the Measurements section, it is believed that the probe measurements exaggerate the radial velocity values in the regions of low axial velocity and peak radial velocity. Consequently, excessive reliance should not be placed on the measured radial velocities. However, the radial velocity values are an indicator of the direction and magnitude of the jet spread. Both the magnitude of the radial velocity and its rate of decay along the furnace increase with swirl. The features shown in Fig. 5 are consistent with the jet expansion behavior suggested by the axial velocity distributions.

Static Pressure Profiles

Static pressure distributions indicate that all pressure values are below ambient pressure, with maximum depressions near the burner. These distributions are similar to those generally observed in enclosed swirling jet flows in furnaces, with greatest depressions at and around the center of the jet and flat pressure distributions within the reverse-flow zones. Increased swirl, above the value corresponding to the onset of the CRZ, increases the centerline depression, resulting in a larger CRZ. The radial gradients of the static pressure are higher in the higher swirl case, and it takes a longer axial distance to reach a uniform pressure distribution in this case.

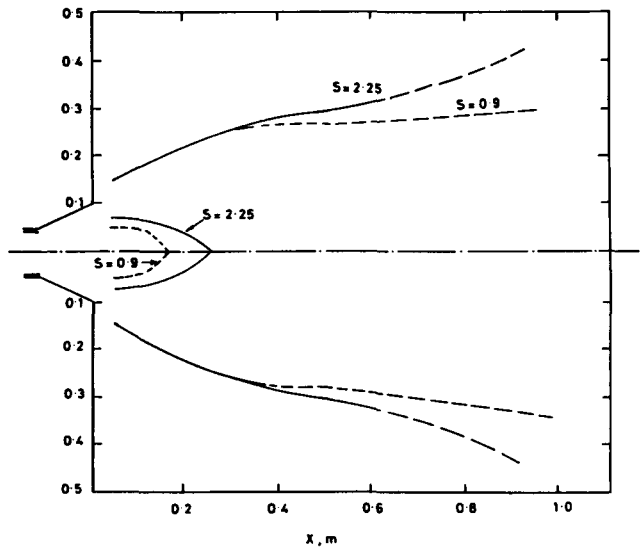


Figure 3. Flow boundaries.

COMBUSTION PATTERN RESULTS

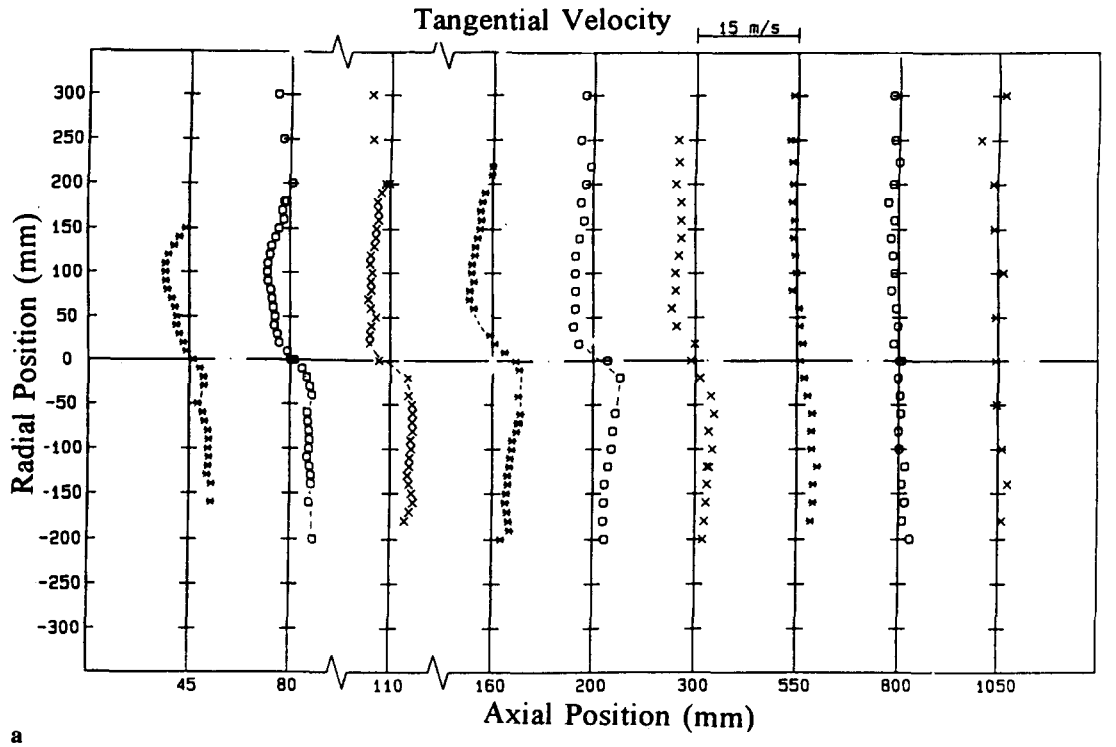
The flames observed were of the short intense type associated with high mixing rates and central reverse flow. Combustion patterns are represented by the distributions of time-averaged temperature and species concentrations within the furnace.

Temperature Distributions

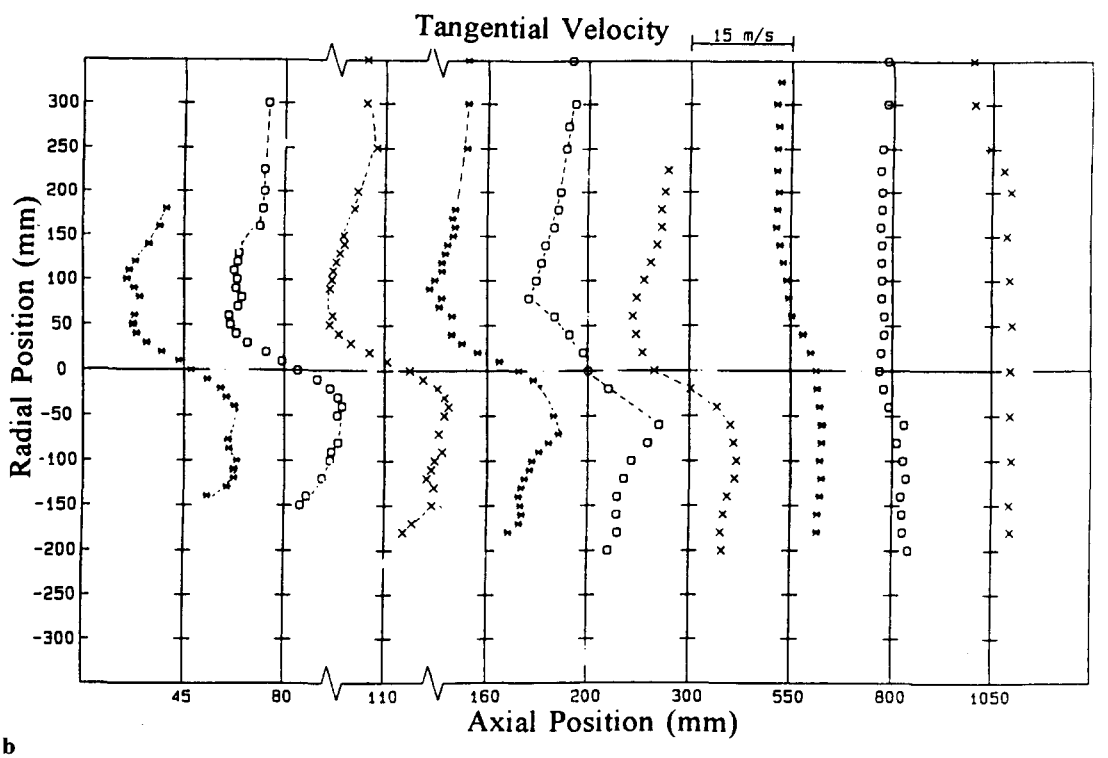
Temperature distributions are presented as contours in Fig. 6. Temperature profiles are also shown in Fig. 7 for three of the axial positions along the furnace. These figures give an indication of the effect of swirl on the rate of combustion as well as the symmetry of the flame. Generally, the profiles shown in Fig. 7 illustrate a good degree of symmetry, particularly for the higher swirl case. The slight asymmetry in the lower swirl case near the burner may be caused by slight variation in the width of the slit used for the fuel gas injection (see also [1]).

The temperature contours indicate that combustion starts within the quarl, a fact confirmed by the measured species concentrations. They also show that the maximum temperature and steepest radial temperature gradients occur nearer to the burner as swirl is increased from 0.9 to 2.25. Further downstream, these gradients decay much more rapidly at the higher swirl. Away from the furnace walls, for the higher swirl case, near-uniform temperature profiles are reached at a plane 1 m downstream of the burner. However, for the lower swirl, radial temperature gradients are evident even at the furnace exit. Thus, increased swirl not only increased combustion intensity but also enhanced the stirring of the post-flame gases.

Within the CRZ, temperature profiles are mostly uni-



a



b

Figure 4. Tangential velocity profiles: (*a*) $S = 0.90$, (*b*) $S = 2.25$.

form for both swirl cases. The temperature at the centerline is always close to the maximum temperature measured at this traverse plane. Within the ERZ, Fig. 7 shows that for both swirl settings the gas temperature decreases from about 1000°C at 545 mm to about 600°C at 45 mm, this being due to heat transfer to the furnace walls.

Main Species Concentrations

Complete surveys of concentrations of HC, CO, CO_2, O_2, and NO_x were performed along the furnace. For brevity, only some radial profiles are presented at two planes close to the burner (Figs. 8–11), where local combustion information is required. This information includes the effect of

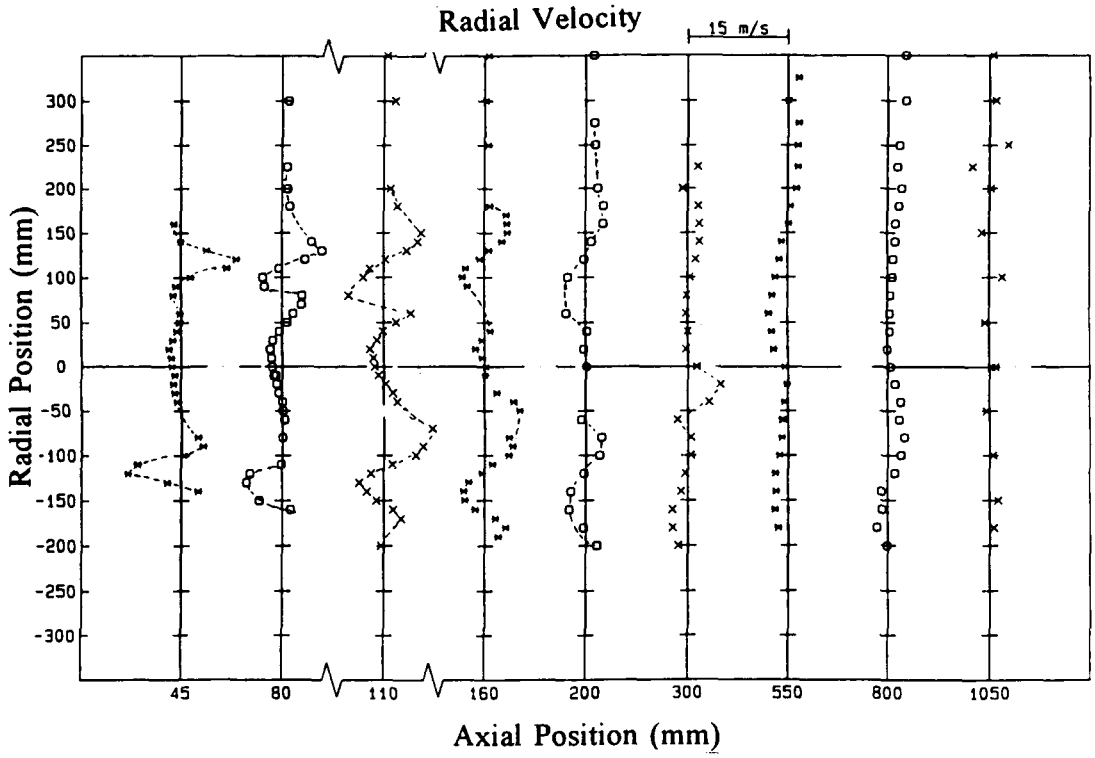


Figure 5. Radial velocity profiles: $S = 2.25$.

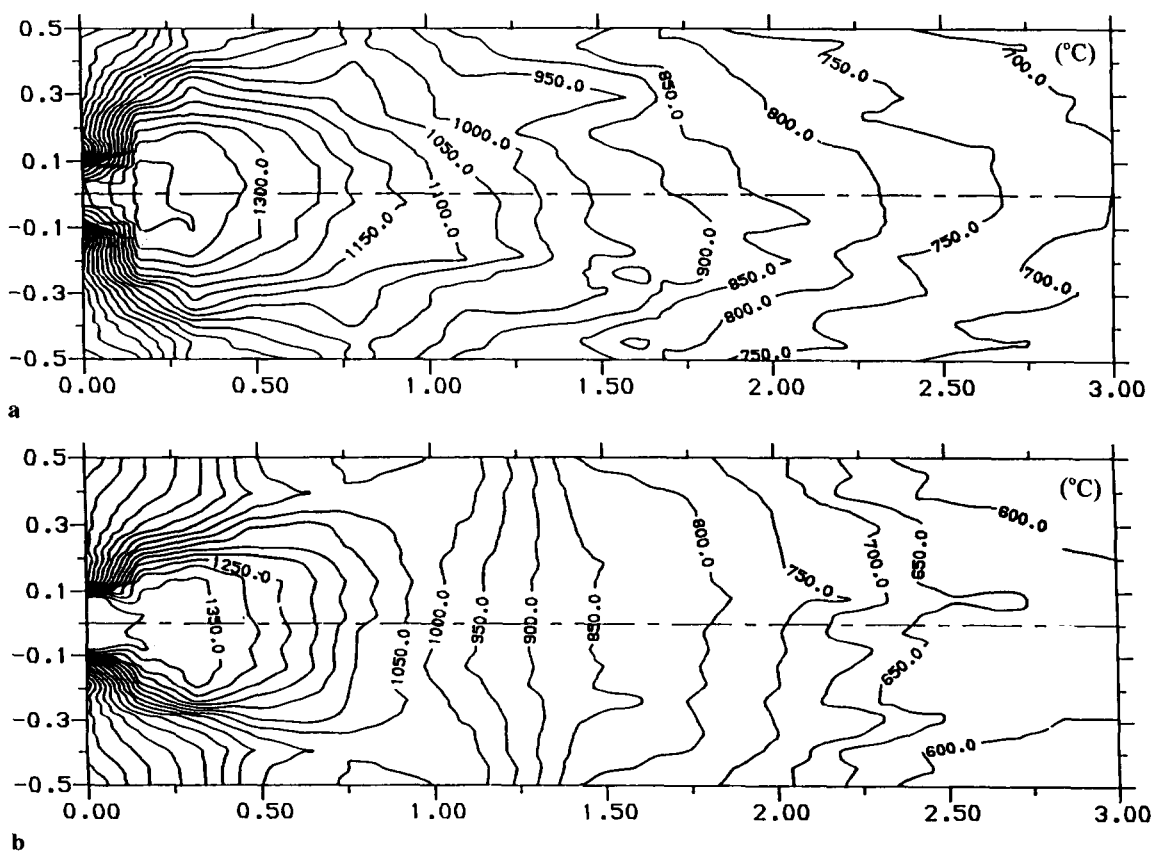


Figure 6. Temperature contours (°C): (*a*) $S = 0.90$, (*b*) $S = 2.25$.

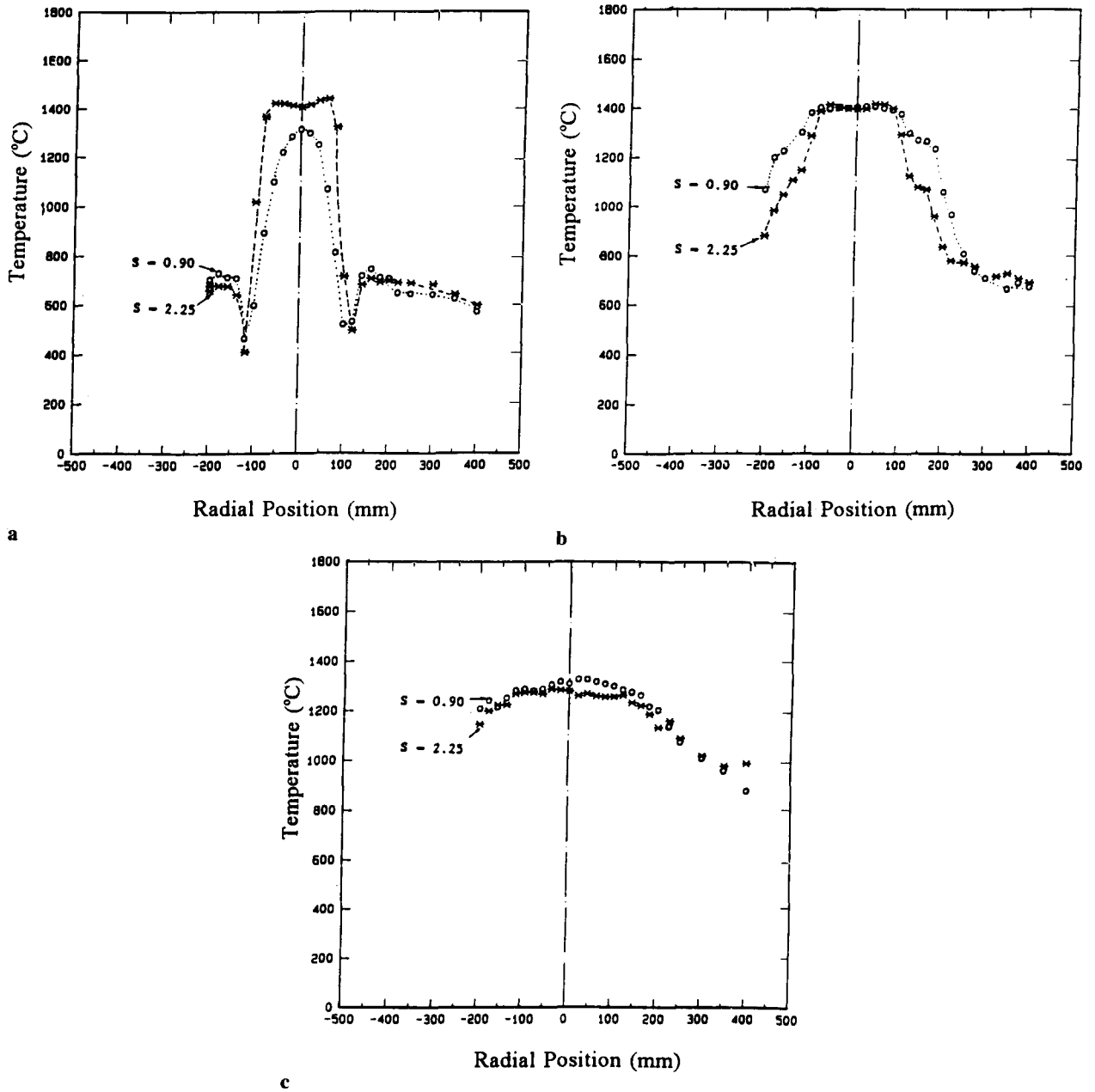


Figure 7. Radial temperature profiles: (*a*) X = 45 mm, (*b*) X = 160 mm, (*c*) X = 545 mm.

swirl and degree of mixing on combustion intensity and flame size.

The concentration profiles shown in Figs. 8–11 exhibit the same degree of symmetry as described for the temperature profiles.

On inspection of the complete profiles, the flow field can be divided into four zones: A, the main reaction zone (near burner forward flow); B, the central reverse-flow zone (CRZ); C, the fully developed flow zone; and D, the external reverse-flow zone (ERZ).

Most of the combustion occurs in the reaction zone (A), leading to uniform concentration distributions in the fully developed flow zone (C), as detailed below.

Zone A In the reaction zone, the concentrations of HC and CO are higher in the case of $S = 0.9$. This shows that the degree of mixing and consequently completeness of combustion are lower. This is also confirmed by the lower measured CO_2 and the higher O_2 in this region. As swirl is increased, high CO_2 peaks are observed in the annular forward jet and correspondingly lower O_2, CO, and HC, consistent with increased combustion intensity. The flame envelope is therefore smaller in the case of $S = 2.25$. From the CO contours it can be estimated that the flame diameter for the higher swirl is about 0.8 that for the lower swirl, in contrast with the wider jet spread.

The minimum value of CO_2 at the first measuring

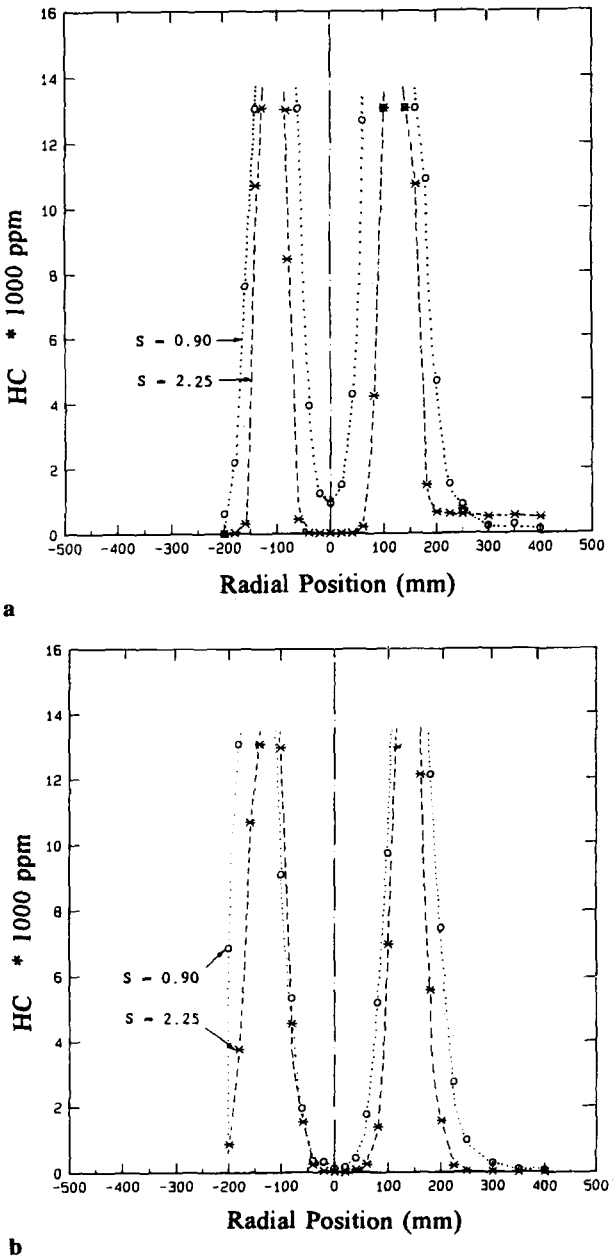


Figure 8. Radial HC profiles: (*a*) X = 45 mm, (*b*) X = 160 mm.

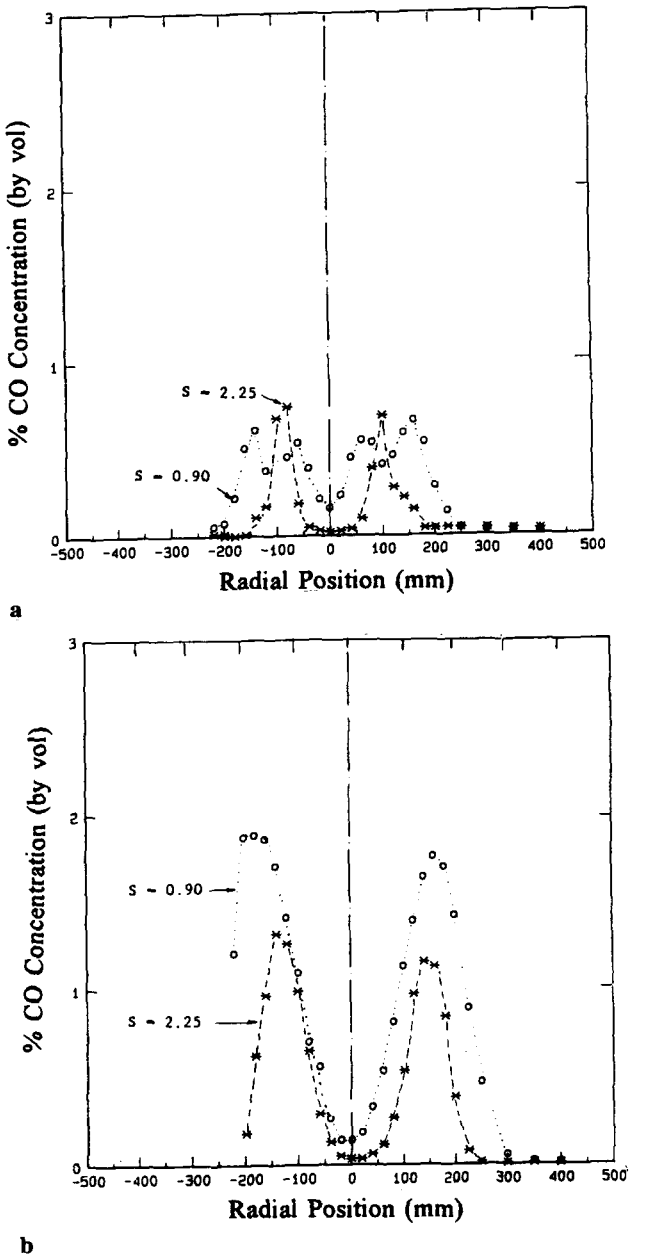


Figure 9. Radial CO profiles: (*a*) X = 45 mm, (*b*) X = 160 mm.

section, 45 mm from the quarl exit (Fig. 10a) varies between 2.5% and 4.0%, confirming that combustion commences within the quarl. This is also confirmed by the maximum value of O_2 concentration at this section. The measurements show that the main reaction zone ends at about 500 mm. Beyond this distance, concentrations of CO and HC decay to zero, and concentrations of CO_2 and O_2 attain their uniform values of 11.5% and 1%, respectively, representing complete combustion.

Zone B Within the CRZ, concentration of HC and CO are nearly zero in the case of $S = 2.25$. However, for $S = 0.90$, small concentrations exist along the CRZ. This indicates that the CRZ of the lower swirl flame receives mixtures with reactions still in progress.

Zone C As mentioned for Zone A, the fully developed flow zone extends from about 500 mm downstream from the quarl to the furnace exit.

Zone D In the ERZ, although gases circulating in this region are mainly products of complete combustion, very low concentrations of CO and HC are observed for both swirls. These concentrations are entrained from the outer boundary of the forward flow where reactions are still in progress.

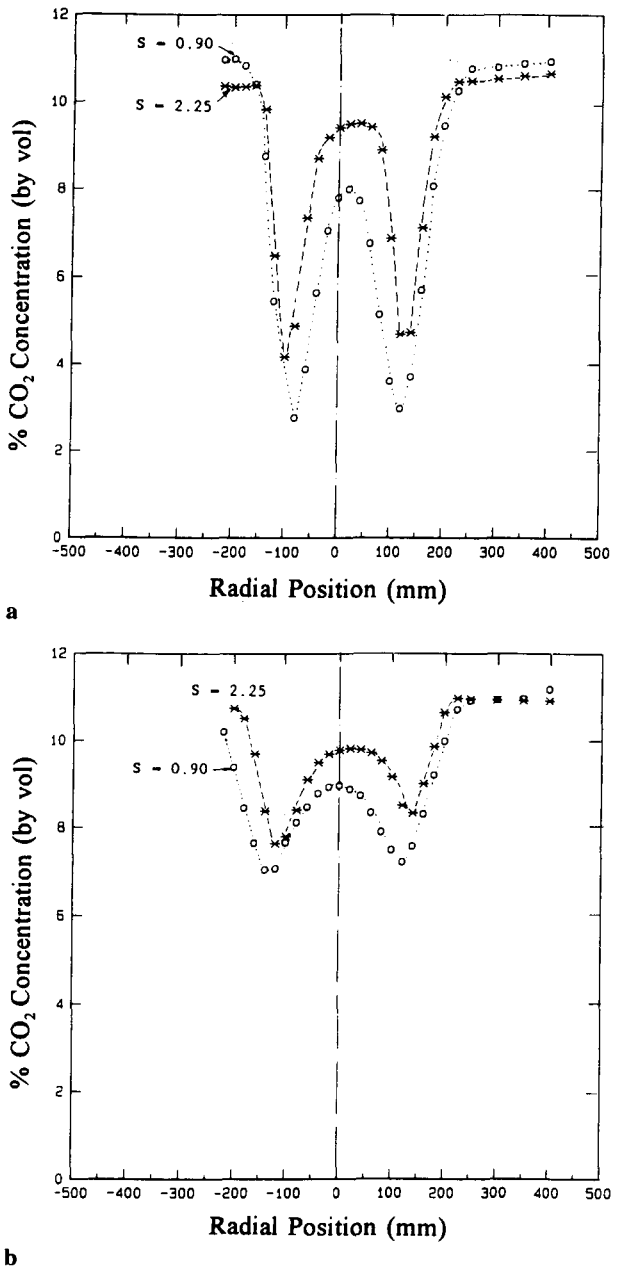


Figure 10. Radial CO_2 profiles: (*a*) X = 45 mm, (*b*) X = 160 mm.

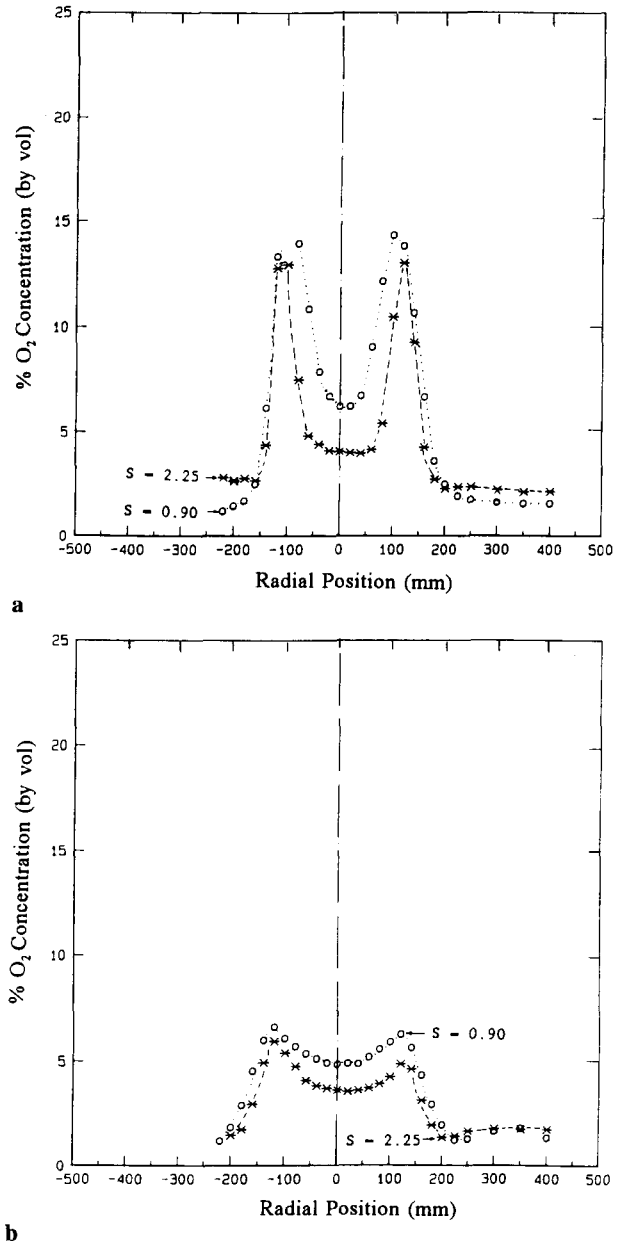


Figure 11. Radial O_2 profiles: (*a*) X = 45 mm, (*b*) X = 160 mm.

NO_x Concentrations

The results of NO_x measurements for the two swirl settings surveyed are represented in Fig. 12 by some illustrative NO_x profiles at near-burner planes. These results are discussed in relation to the temperature and species concentration fields described above, using the same four flow-field zones.

Zone A The reaction zone is characterized by steep gradients of NO_x concentration in both the radial and axial directions. The rate of NO_x formation follows closely the rate of the main combustion reactions as indicated by the temperature and main species concentrations. Comparison of the NO_x profiles at 45 mm from the quarl exit (Fig. 12) with those of the temperature (Fig. 7) shows that NO_x profiles are similar to the temperature profiles for each swirl case. At the 545-mm plane from the quarl exit, NO_x profiles become almost uniform across the furnace. These features indicate that in this case NO_x is formed by the thermal mechanism. However, it is noted that the NO_x concentration levels are significantly higher for the lower swirl case. This cannot be completely explained by the temperature variation alone. Two other factors must also be considered.

1. Longer residence time, due to the slower reaction
2. Higher fluctuations in the concentrations as observed from the flame shape.

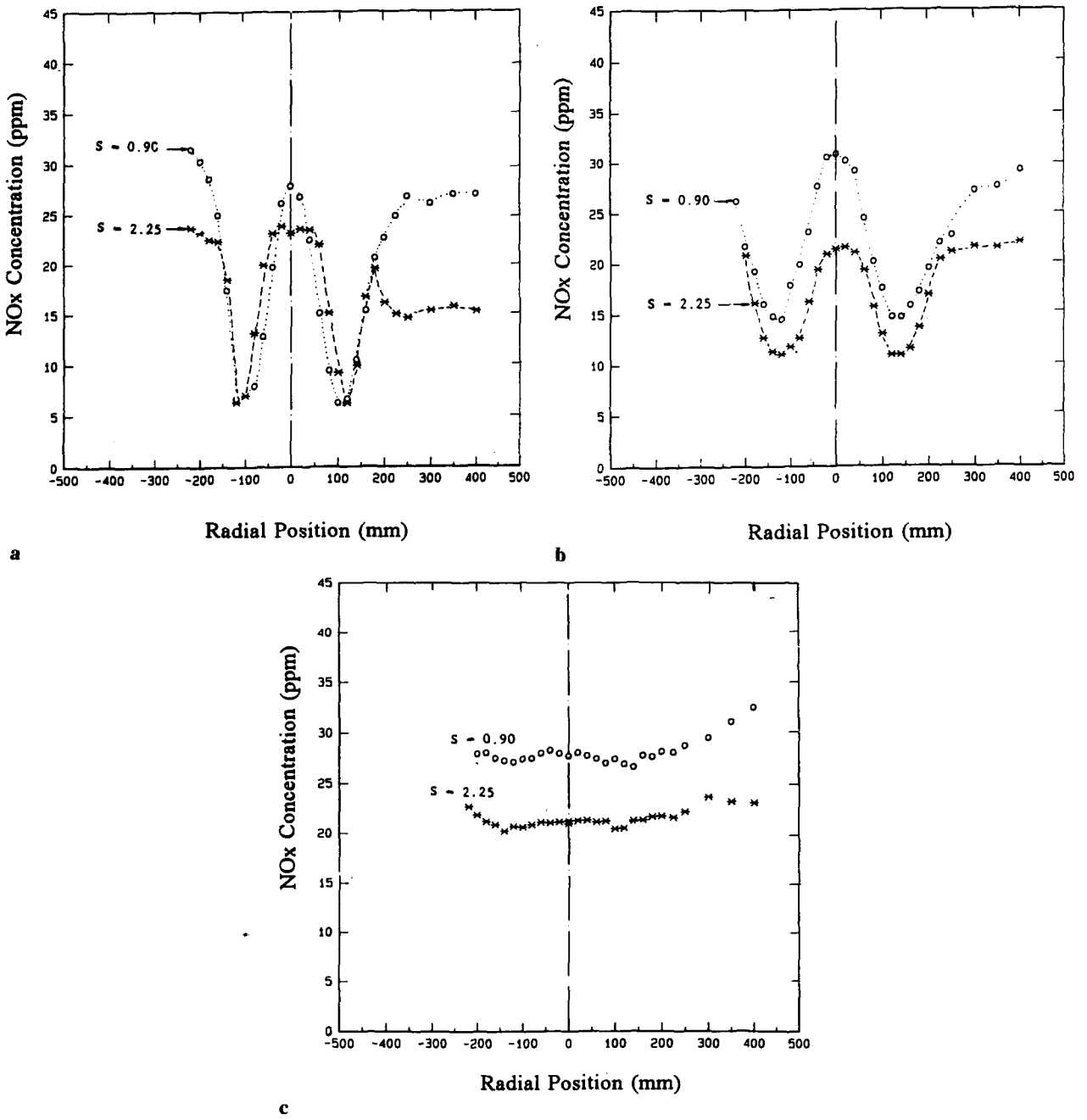


Figure 12. Radial NO_x profiles: (*a*) X = 45 mm, (*b*) X = 160 mm, (*c*) X = 545 mm.

Zone B The profiles of NO_x within the CRZ (Fig. 12) and the corresponding temperature profiles (Fig. 7) shows that NO_x concentrations within this region are due mainly to the entrainment from the forward flow and not due to reactions within this zone. For example, at the 45-mm plane, higher NO_x concentrations are measured for the lower swirl although the temperature is lower. Further downstream, at the 160-mm plane, the temperature levels at both swirl settings are about the same, yet higher levels of NO_x are observed at the lower swirl.

Zone C At both swirl settings, constant NO_x concentrations were measured throughout this region, and these equal the value measured at the stack. The lower swirl, $S = 0.9$, produced the higher NO_x value of 32 ppm compared to only 26 ppm at $S = 2.25$.

Zone D The flow in the ERZ comes mainly from the fully developed flow with some flow entrained from the outer boundary of the reaction zone. Figure 12 shows that NO_x concentration within this zone increases with axial distance from the burner until it reaches the fully developed value.

The discussion of NO_x results above confirms that the formation of NO_x in these flames is mainly by the

Zeldovich mechanism, although there may be little prompt NO_x. The formation rates are affected by the flame temperature, residence time of hot gases in reaction region, and concentration fluctuations. The last two of these factors lead to higher NO_x values in the lower swirl case.

It should be noted that the measurements reported in this paper were taken in a large furnace that was firing under a fairly moderate load. If such a furnace were fired more heavily with the same fuel, it would be expected that the temperature patterns in the NO_x formation zones would remain roughly the same. Residence times would be reduced. Mixing distributions would, however, possibly alter, with more significant fluctuations. Further experimental work is needed to study the effect of furnace loading on pollutant formation.

PRACTICAL SIGNIFICANCE

This paper offers burner designers a nonconventional mode of fuel injection and demonstrates its advantages. The results presented constitute detailed data of flow and combustion patterns. These data should help designers optimize combustion performance with reduced pollutant emissions. In addition to providing more information and insight into the processes involved, the detailed data obtained will also be used for the validation and development of the mathematical models used for furnace design.

CONCLUSIONS

1. Using the air entry scheme 1 with the peripheral injection system, a minimum swirl ($S = 0.8$) was required to achieve a CRZ stabilized flame. In scheme 2, a CRZ was created by introducing some of the combustion air radially outward through a central gun; thus a stable flame was established even without swirl.
2. The flow patterns presented for scheme 1 with peripheral injection were represented by distributions of the three time-averaged velocity components and the static pressure. These measurements have given clear pictures of flow expansion in the radial and axial directions as well as of the flow boundaries and thus the central and external reverse-flow zones (CRZ and ERZ). The effect of swirl on the size of the CRZ was found to be significant with this injection system due to the increased influence of centrifugal forces.
3. An assessment of the combustion patterns was obtained from measurements of temperature and species concentrations. Good flame symmetry was obtained. Increasing swirl enhanced the mixing and thus combustion intensity.
4. Measurements of the local NO_x concentrations showed that NO_x formation takes place through the thermal mechanism, with strong dependence on local flame temperature and O_2 availability. The NO_x formation rate is thus dependent on the degree of mixing and local concentration fluctuations. In addition it depends on the residence time within the reaction zone. Consequently, at the higher swirl, higher mixing led to reduced NO_x formation.

Part of this work was carried out under the research program of Heat Transfer and Fluid Flow Service (HTFS), which receives support from the U.K. Department of Trade and Industry.

REFERENCES

1. Beltagui, S. A., and Maccallum, N. R. L., Characteristics of Enclosed Swirl Flames with Peripheral Fuel Injection, *J. Inst. Energy* **61**, 3–16, 1988.
2. Beltagui, S. A., Kenbar, A. M. A., and Maccallum, N. R. L., Comparison of Measured Isothermal and Combusting Confined Swirling Flows—Peripheral Fuel Injection, Proc. 2nd World Conference on Experimental Heat Transfer, Fluid Mechanics and Thermodynamics, pp. 118–125, Elsevier, New York, 1991.
3. Kenbar, A. M. A., Beltagui, S. A., and Maccallum, N. R. L., Effect of Fuel Injection Modes on the Combustion Pattern and Pollution Formation in a Gas Fired Furnace, 2nd Int. Conf. on Combustion Technologies for a Clean Environment, Lisbon, Portugal, Paper 26.3, July 1993.
4. Beltagui, S. A., Fuggle, R. N., Kenbar, A. M. A., Ralston, T., Marriott, N., and Stopford, P. J., Modelling a Gas-Fired Furnace Flow and Combustion Using the PCOC Code, Euroteck Direct 91 Conf., I. Mech. Eng., pp. 51–58, Birmingham, July 1991.
5. Kenbar, A. M. A., Beltagui, S. A., and Maccallum, N. R. L., Modelling the Combustion Aerodynamics for a Peripheral Injection Flow, 5th Int. PHOENICS Users Conf., Nice, France, September 1992.
6. Kenbar, A. M. A., Beltagui, S. A., Ralston, T., and Maccallum, N. R. L., Measurement and Modelling of NO_x Formation in a Gas Fired Furnace, 1st Int. Conf. on Combustion Technologies for a Clean Environment, Vilamoura, Portugal, Paper 19.2, September 1991.
7. Kenbar, A. M. A., Combustion Aerodynamics and Pollutant Formation in Gas-Fired Furnaces, Ph.D. Thesis, Mech. Eng. Dept., Univ. Glasgow, 1991.
8. Chedaille, J., and Braud, Y., *Measurements in Flames*, Edward Arnold, London, 1972.
9. Hillemanns, R., Lenze, B., and Leuckel, W., Flame Stabilization and Turbulent Exchange in Strongly Swirling Natural Gas Flames, 21st Symp. (Int.) on Combustion, The Combustion Institute, Pittsburgh, PA, pp. 1445–1453, 1987.
10. Beltagui, S. A., and Maccallum, N. R. L., Aerodynamics of Vane-Swirled Flames in Furnaces, *J. Inst. Fuel* **49**, 183–193, 1976.

Received February 15, 1994; revised September 15, 1994.